

AMENDMENT NO. 1 AUGUST 2017
TO
IS 6911 : 2017 STAINLESS STEEL PLATE, SHEET AND STRIP —
SPECIFICATION

(Second Revision)

(Page 3, Table 1) — Substitute the following for the existing table:

See Table 1 on page 2.

[Page 6, Table 3, SI No. (i) (3), col 2] — Substitute ‘X02Cr12Ti’ *for* ‘X02Cr12TiNb.’

[Page 6, Table 3, SI No. (i) (5), col 2] — Substitute ‘X02Cr12’ *for* ‘X02Cr12Ni.’

[Page 6, Table 3, SI No. (iii) (12)] — Insert the following new entry and renumber the subsequent serial numbers:

SI No.	Grade Designation	Symbols*	Annealing or Softening Temperature °C	Quenching Media for Annealing or Softening ¹⁾	Symbols ²⁾	Hardening Temperature °C	Quenching Media for Hardening	Tempering Temperature °C	
	Letter Symbols [see IS 1762 (Part 1)]	Numerical Symbol ISS							
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Austenitic Steels									
13)	X06Cr18Ni12	305	SA	1 000-1 120	w, a	-	-	-	-
14)	X04Cr19Ni10Cu3Mo	305 Cu	SA	1 000-1 120	w, a	-	-	-	-

(Page 7, clause 10.4) — Substitute the following for the existing clause:

‘10.4 The material supplied in annealed condition and tested as per **10.1**, **10.2** and **10.3** and shall conform to the requirements of Table 4.’

(Page 8, Table 4) — Substitute the following for the existing table:

See Table 4 on page 5.

Table 1 Chemical Composition, Percent
(Clauses 7.1 and 7.2)

Sl No.	Grade Designation	Numerical Symbol ISS	C, Max	Si, Max	Mn	Ni	Cr	Mo	S, Max	P, Max	N	Others
	Letter Symbol [see IS 1762 (Part 1)] (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
i)	<i>Ferritic steels:</i>											
1	X 04Cr12Al	405	0.08 Max	1.00	1.00 Max	0.60 Max	11.5-14.5	–	0.030	0.040	–	Al 0.10-0.30
2	X 07Cr17	430	0.12 Max	1.00	1.00 Max	0.75 Max	16.0-18.0	–	0.030	0.040	–	–
3	X 02Cr12Ti	409	0.030 Max	1.00	1.00 Max	0.50 Max	10.5–11.7	–	0.020	0.040	0.030 Max	Ti 6x (C+N) Min—0.50 Max; Nb 0.17 Max
4	X 02Cr12	409 M	0.030 Max	1.00	0.50-1.50	1.50 Max	10.5-12.5	–	0.030	0.040	–	–
5	X 04Cr12	410S	0.08 Max	1.00	1.00 Max	0.60 Max	11.5–13.5	–	0.030	0.040	-	
6	X 02Cr18Ti	439	0.030 Max	1.00	1.00 Max	0.50 Max	17.0-19.0	-	0.030	0.040	0.030 Max	Ti [0.2+4 (C+N)] Min— 1.1 Max; Al 0.15 Max
ii)	<i>Martensitic steels:</i>										–	–
1	X 12Cr12	410	0.08-0.15	1.00	1.00 Max	0.75 Max	11.5-13.5	–	0.030	0.040	–	–
2	X 20Cr13	420 S1	0.16-0.25	1.00	1.00 Max	1.00 Max	12.0-14.0	-	0.030	0.040	–	–
3	X 30Cr13	420 S2	0.26-0.35	1.00	1.00 Max	1.00 Max	12.0-14.0	–	0.030	0.040	–	–
4	X 40Cr13	420S3	0.35-0.45	1.00	1.00 Max	1.00 Max	12.5-14.0	–	0.030	0.040	–	–
5	X 15Cr16Ni2	431	0.10-0.20	1.00	1.00 Max	1.25-2.50	15.0-17.0	–	0.030	0.045	–	–
6	X 108Cr17Mo	440	0.95-1.20	1.00	1.00 Max	0.50 Max	16.0-18.0	0.75 Max	0.030	0.045	-	-
iii)	<i>Austenitic steels:</i>											–
1	X 10Cr17Mn6Ni4N20	201	0.15 Max	1.00	5.5-7.5	3.5-5.5	16.0-18.0	-	0.030	0.060	0.25 Max	–
2	X 07Cr17Mn12Ni4	201 A	0.12 Max	1.00	10.0-14.0	3.5-5.5	16.0.18.0	-	0.030	0.090	0.25 Max	–
3	X 10Cr18Mn9Ni5	202	0.15 Max	1.00	7.5-10.0	4.0-6.0	17.0-19.0	–	0.030	0.060	0.25 Max	–
4	X 10Cr17Ni7	301	0.15 Max	1.00	2.00 Max	6.0-8.0	16.0-18.0	–	0.030	0.045	0.10 Max	–
5	X 07Cr18Ni9	302	0.15 Max	0.75	2.00 Max	8.0-10.0	17.0-19.0	–	0.030	0.045	0.10 Max	–
6	X 02Cr17Ni7	301L	0.03 Max	1.00	2.00 Max	6.0-8.0	16.0–18.0	–	0.030	0.045	0.20 Max	
7	X02Cr17Ni7N	301LN	0.03 Max	1.00	2.00 Max	6.0-8.0	16.0–18.0	–	0.030	0.045	0.07-0.20	-
8	X04Cr19Ni9	304 S1	0.07 Max	0.75	2.00 Max	8.0-10.5	17.5-19.5	–	0.030	0.045	0.10 Max	–
9	X 02Cr19Ni10	304 S2	0.030 Max	0.75	2.00Max	8.0.12.0	17.5-19.5		0.030	0.045	0.10 Max	–

Table 1 (Continued)

Sl No.	Grade Designation	Numerical Symbol ISS	C, Max	Si, Max	Mn	Ni	Cr	Mo	S, Max	P, Max	N	Others
	Letter Symbol [see IS 1762 (Part 1)]											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
10	X 07Cr19Ni9	304H	0.04-0.10	0.75	2.00 Max	8.0-10.5	18.0-20.0	—	0.030	0.045	-	—
11	X 02Cr19Ni10N	304LN	0.030 Max	0.75	2.00 Max	8.0-12.0	18.0-20.0	—	0.030	0.045	-	—
12	X 04Cr19Ni9N	304N	0.08 Max	0.75	2.00Max	8.0-10.5	18.0-20.0	—	0.030	0.045	-	-
13	X06Cr18Ni12	305	0.12Max	0.75	2.00 Max	10.5-13.0	17.0-19.0	—	0.030	0.045	-	-
14	X04Cr19Ni10Cu3Mo	305 Cu	0.08Max	0.50-2.50	2.00Max	8.5-11.5	17.0-20.5	0.75-1.50	0.030	0.045	-	Cu 0.75-3.50
15	X15Cr24Ni13	309	0.20 Max	1.5	2.00 Max	11.0-15.0	22.0-25.0	—	0.030	0.045	—	—
16	X 04Cr23Ni14	309S	0.08 Max	0.75	2.00 Max	12.0-15.0	22.0-24.0	—	0.030	0.045	-	-
17	X20Cr25Ni20	310	0.25Max	2.5	2.00 Max	18.0-21.0	24.0-26.0	—	0.030	0.045	—	—
18	X 04Cr25Ni20	310S	0.08 Max	1.50	2.00 Max	19.0-22.0	24.0-26.0	—	0.030	0.045	-	-
19	X 04Cr17Ni12Mo2	316	0.08 Max	0.75	2.00 Max	10.0-14.0	16.0-18.0	2.0-3.0	0.030	0.045	0.10 Max	—
20	X 02Cr17Ni12Mo2	316 L	0.030 Max	0.75	2.00 Max	10.0-14.0	16.0-18.0	2.0-3.0	0.030	0.045	0.10 Max	—
21	X 07Cr17Ni12Mo2	316H	0.04-0.10	0.75	2.00 Max	10.0-14.0	16.0-18.0	2.0-3.0	0.030	0.045	—	—
22	X 02Cr17Ni12Mo2N	316LN	0.030 Max	0.75	2.00 Max	10.0-14.0	16.0-18.0	2.0-3.0	0.030	0.045	0.10-0.16	—
23	X 04Cr19Ni13Mo3	317	0.08 Max	0.75	2.00 Max	11.0-15.0	18.0-20.0	3.0-4.0	0.030	0.045	0.10	—
24	X 02Cr19Ni13Mo3	317L	0.030 Max	0.75	2.00Max	11.0-15.0	18.0-20.0	3.0-4.0	0.030	0.045	0.10 Max	—
25	X 04Cr17Ni12Mo2Ti	316 Ti	0.08 Max	0.75	2.00 Max	10.0-14.0	16.0-18.0	2.0-3.0	0.030	0.045	0.10 Max	Ti 5x (C + N) Min— 0.70 Max
26	X 04Cr18Ni10Ti	321	0.08 Max	0.75	2.00 Max	9.0-12.0	17.0-19.0	—	0.030	0.045	0.10 Max	Ti 5 x(C + N) Min— 0.70 Max
27	X 04Cr18Ni10Nb	347	0.08 Max	0.75	2.00 Max	9.0-13.0	17.0-19.0	—	0.030	0.045	—	Nb 10xC – 1.00 Max
28	X 10Cr15Mn9Cu2Ni1N	N1	0.12 Max	0.75	8.5-10.5	1.0-2.0	14.5-16.0	—	0.030	0.080	0.08-0.20	Cu 1.5-2.50
29	X 8Cr16Mn8Cu2Ni2N	N2	0.10 Max	1.0	6.5-9.0	1.5-3.5	15.5-17.0	—	0.030	0.070	0.10-0.25	Cu 2.0-4.0
30	X 8Cr16Mn7Cu2Ni4N	N3	0.09 Max	0.75	6.0-8.0	4.0-6.0	16.0-17.5	—	0.030	0.070	0.05-0.15	Cu 1.5-2.50
iv)	Duplex:											
1	X 02Cr22Ni6Mo3N	2205	0.030 Max	1.00	2.00Max	4.5-6.5	22.0-23.0	3.0-3.5	0.020	0.03	0.14-0.20	-
2	X 02Cr23Ni4CuN	2304	0.030 Max	1.00	2.50Max	3.0-5.5	21.5-24.5	0.05-0.60	0.030	0.04	0.05-0.20	Cu 0.50-0.60
3	X 02Cr25Ni7Mo4CuN	2507	0.030 Max	0.80	1.20Max	6.0-8.0	24.0-26.0	3.0-5.0	0.020	0.035	0.24-0.32	Cu 0.50 Max

Table 1 (Concluded)

Sl No.	Grade Designation	Numerical Symbol ISS	C, Max	Si, Max	Mn	Ni	Cr	Mo	S, Max	P, Max	N	Others
	Letter Symbol [see IS 1762 (Part 1)] (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Constituent			Limits, Percent, Max									
			Ferritic and Martensitic Steels					Austenitic Steels				
								Without Specified Molybdenum		With Specified Molybdenum		
Titanium			—					0.10		0.10		
Niobium			—					0.20		0.20		
Molybdenum			0.30					0.70		—		
Copper			0.30					0.50		0.70		

Table 4 Mechanical Properties in Annealed Condition
(Clauses 10.3.1 and 10.4)

Sl No.	Grade Designation		Hardness, <i>Max</i>			Yield Strength, <i>M_{in}</i> , MPa 0.2 Percent Proof Stress	Tensile Strength <i>M_{in}</i> , MPa	Elongation Percent, <i>M_{in}</i> 50 mm	Bend Test D _{ma}
	Letter Symbol [see IS 1762 (Part 1)]	Numerical Symbol ISS	Brinell HBS/HBW	Rockwell HRB					
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
1	X04Cr12Al	405	179	88	170	415	20	2a*	
2	X07Cr17	430	183	89	205	450	22	2a*	
3	X02Cr12Ti	409	179	88	170	380	20	a*	
4	X02Cr12	409 M	180	88	280	450	18	a*	
5	X04Cr12	410 S	183	89	205	415	22	a*	
6	X02Cr18Ti	439	183	89	205	415	22	a*	
7	X12Cr12	410	217	96	205	450	20	a*	Not required
8	X20Cr13	420 S1	225	95	-	700, <i>Max</i>	15		Not required
9	X30Cr13	420 S2	235	97	-	740, <i>Max</i>	15		Not required
10	X40Cr13	420 S3	240	98	-	760, <i>Max</i>	12		Not required
11	X15Cr16Ni2	431	285	29 ^A	-	-	-		Not required
12	X10Cr17Mn6Ni4N20	201	217	95	260	515	40		Not required
13	X07Cr17Mn12Ni4	201 A	217	95	260	540	40		Not required
14	X10Cr18Mn9Ni5	202	241	100	260	620	40		Not required
15	X10Cr17Ni7	301	217	95	205	515	40		Not required
16	X07Cr18Ni9	302	201	92	205	515	40		Not required
17	X02Cr17Ni7	301 L	241	100	220	550	45		Not required
18	X02Cr17Ni7N	301 LN	241	100	240	550	45		Not required
19	X04Cr19Ni9	304 S1	201	92	205	515	40		Not required
20	X02Cr19Ni10	304 S2	201	92	170	485	40		Not required
21	X07Cr19Ni9	304 H	201	92	205	515	40		Not required
22	X02Cr19Ni10N	304 LN	217	95	205	515	40		Not required
23	X04Cr19Ni9N	304 N	217	95	240	550	30		Not required
24	X06Cr18Ni12	305	183	88	170	485	40		Not required
25	X04Cr19Ni10Cu3Mo	305 Cu	201	92	205	515	40		Not required
26	X15Cr24Ni13	309	217	95	210	490	40		Not required
27	X04Cr23Ni14	309 S	217	95	205	515	40		Not required
28	X20Cr25Ni20	310	217	95	210	490	40		Not required
29	X04Cr25Ni20	310 S	217	95	205	515	40		Not required
30	X04Cr17Ni12Mo2	316	217	95	205	515	40		Not required
31	X02Cr17Ni12Mo2	316 L	217	95	170	485	40		Not required
32	X07Cr17Ni12Mo2	316 H	217	95	205	515	40		Not required
33	X02Cr17Ni12Mo2N	316 LN	217	95	205	515	40		Not required
34	X04Cr19Ni13Mo3	317	217	95	205	515	35		Not required
35	X02Cr19Ni13Mo3	317 L	217	95	205	515	40		Not required
36	X04Cr17Ni12Mo2Ti	316 Ti	217	95	205	515	40		Not required
37	X04Cr18Ni10Ti	321	217	95	205	515	40		Not required
38	X04Cr18Ni10Nb	347	201	92	205	515	40		Not required
39	X10Cr15Mn9Cu2Ni1N	N1	241	100	345	650	40		Not required
40	X8Cr16Mn8Cu2Ni2N	N2	241	100	310	650	40		Not required
41	X8Cr16Mn7Cu2Ni4N	N3	241	100	275	600	40		Not required
42	X02Cr22Ni6Mo3N	2205	293	32 ^A	450	655	25		Not required
43	X02Cr23Ni4CuN	2304	290	32 ^A	400	600	25		Not required
44	X02Cr25Ni7Mo4CuN	2507	310	32 ^A	550	795	15		Not required

a * = thickness of the test piece.
A = Rc scale.

(MTD 16)